

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	12 July 2026
Team ID	Buyya Vaishnavi
Project Name	ShopEZ
Maximum Marks	4 Marks

Technical Architecture:

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Customer, Seller and Admin interaction screens	React 19, TypeScript, Tailwind CSS
2.	Application Logic-1	Authentication and Authorization	Node.js, Express.js
3.	Application Logic-2	Product and Cart Management	Node.js, Express.js
4.	Application Logic-3	Order Placement & Status Management	Node.js, Express.js
5.	Database	Store users, products, carts and orders	MongoDB
6.	Cloud Database	Cloud-hosted database service	MongoDB Atlas
7.	File Storage	Product image URL storage	MongoDB Atlas Storage
8.	External API-1	REST API communication between frontend and backend	Axios
9.	External API-2	Authentication Token Verification	JWT (JSON Web Token)
10.	Machine Learning Model	Not Applicable	N/A
11.	Infrastructure (Server / Cloud)	Frontend and Backend deployment environment	Local System, MongoDB Atlas Cloud

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Frameworks used for application development	React.js, Node.js, Express.js, MongoDB
2.	Security Implementations	Authentication, Authorization and Password Protection	JWT Authentication, bcryptjs Password Hashing
3.	Scalable Architecture	Three-tier architecture with separate frontend, backend and database layers	MERN Stack Architecture
4.	Availability	Users can access services anytime through the web application	React + Node.js + MongoDB Atlas
5.	Performance	Fast product search, login and checkout with optimized API calls	Axios, Express.js, MongoDB Indexing